

SRIKUMARAN VM

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Career Objective

Aspiring Data Scientist and Generative AI Engineer with a strong foundation in Python, Machine Learning, Deep Learning, and NLP. Seeking a challenging role in a progressive organization to leverage my analytical, problem-solving, and AI model development skills to build innovative, data-driven, and AI-powered solutions. Eager to contribute to projects involving predictive modeling, generative AI, and advanced analytics while continuously learning and applying cutting-edge AI technologies. Passionate about transforming complex data into actionable insights to drive strategic decision-making. Committed to collaborating with cross-functional teams to deliver scalable, high-impact AI solutions.

Education

Sacred Heart Convent Anglo Indian Hr. Sec. Schoo.Villupuram Apr 2020 – May 2021
HSC

Paavai College of Engineering, Namakkal, Tamil Nadu Aug 2021 – May 2025
Bachelor of Artificial Intelligence and Data Science Engineering

Technical Skills

Programming Languages: Python, SQL

Libraries Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn

Data Science ML: Machine Learning, Deep Learning, NLP

Tools Visualization: Power BI, GitHub

Projects & Certifications

Intelligent Machine Learning Pipeline Web Application

- Built a Flask-based web app and CLI system for automated ML pipeline generation, including data preprocessing, visualization, model training, and prediction.
- Integrated model evaluation and reporting using Python (Pandas, Scikit-learn, Matplotlib).
- Designed a modular, OOP-driven architecture enabling scalable, reusable, and end-to-end machine learning workflows.

Blockchain-based Smart Contracts for Healthcare Supply Chain

- Developed prototype for tracking and verifying medicine authenticity using Ethereum smart contracts.
- Designed blockchain-based model to ensure transparency and traceability in healthcare supply chain.

Blood Cancer Detection using VGG19 (Deep Learning Project)

- Developed a deep learning model using the VGG19 architecture to classify blood smear images for cancer detection.
- Preprocessed image data, applied transfer learning, and fine-tuned the model for high accuracy and robust evaluation.

Neural Machine Translation (English–Spanish) — PyTorch, TensorFlow, NLP

- Built a Transformer-based seq2seq model (512-dim, 6 layers, 8 heads) with custom attention and optimized hyperparameters.
- Enhanced training with preprocessing, Focal Loss, label smoothing, gradient clipping, and mixed-precision training.
- Developed beam search–based inference for real-time, fluent English–Spanish translations.

Multi-Tenant RAG Chatbot System

- Built a secure, multi-tenant Retrieval-Augmented Generation (RAG) chatbot using Flask, SocketIO, LangChain, and SQLite, supporting user authentication, chat history, document management, and session tracking.
- Integrated optimized vector-based search, RAG prompts, and optional SerpAPI web search for real-time, context-aware responses with asynchronous document processing for high performance.